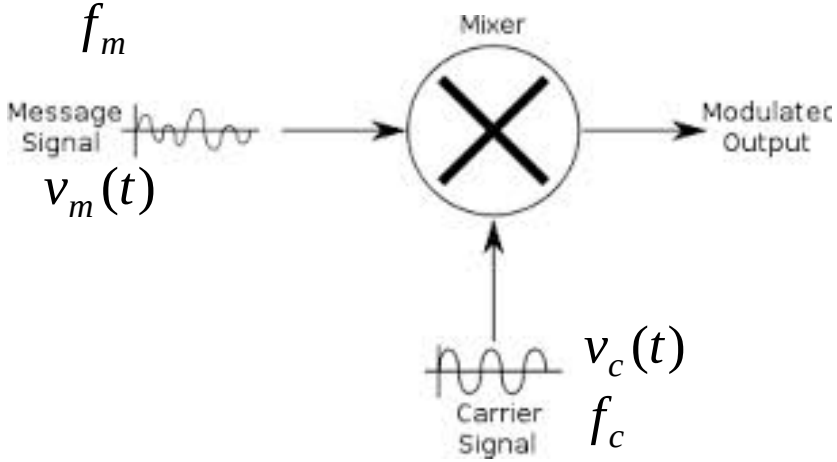
Analog Modulation Schemes

EE3408

Modulation

- Modulation
  - carrier signal $v_c(t)$ is modulated by an information signal $v_m(t)$.

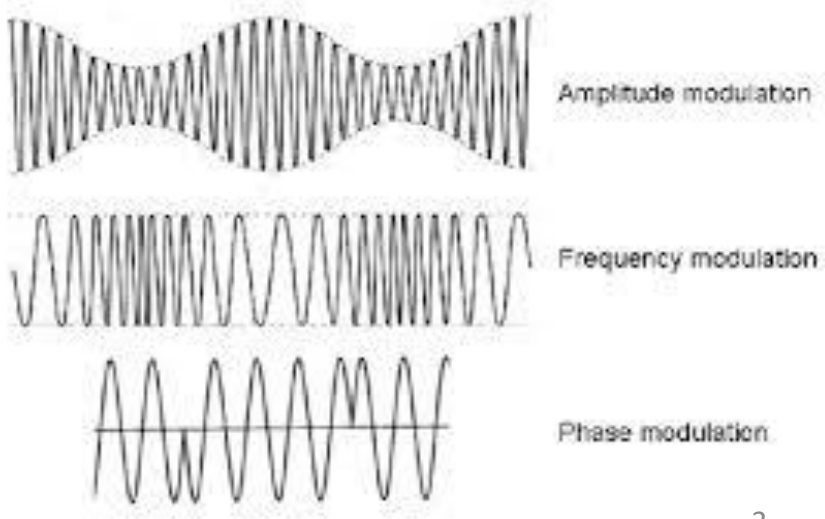


Modulation

- Sinusoidal carrier

$$v_c(t) = V_c \cos(\omega_c t + \emptyset_c)$$

- Parameters of $v_c(t)$ varied in relation with $v_m(t)$
  - Amplitude varied = AM
  - Frequency varied = FM
  - phase varied = PM



Modulation

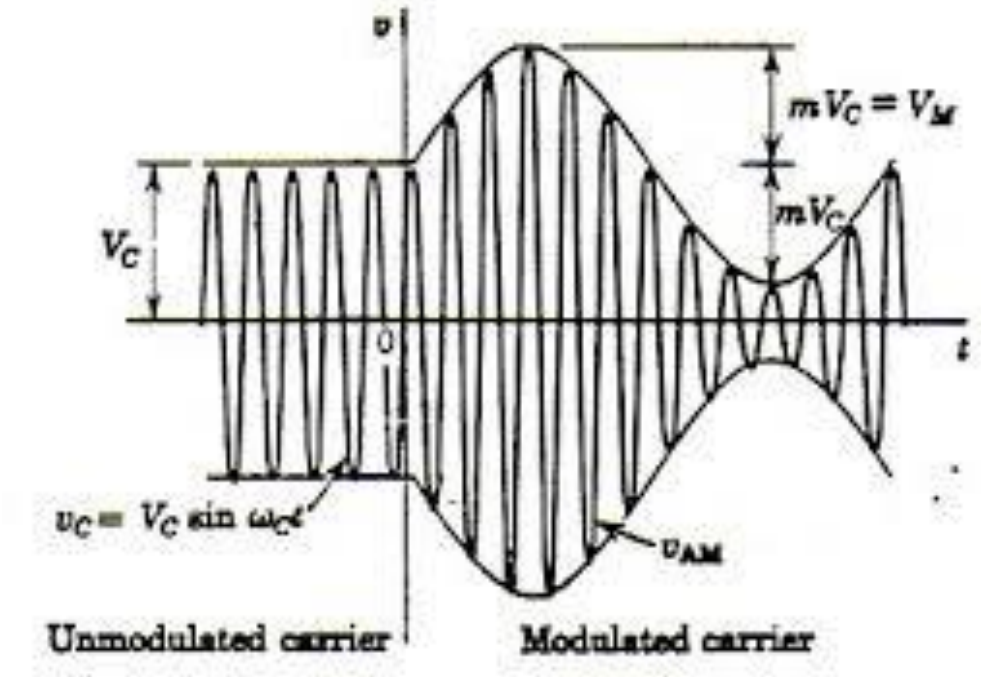
- AM (or envelope modulation)

$$V_c \ \alpha \ V_m(t)$$

- Modulated $v_c(t)$ becomes

$$V_{am}(t) = \ [kv_m(t) + \ C]\, cos(\omega_c t + \ \emptyset_c)$$

- AM – forms
  - DSBC or DSB-WC or DSB-LC
  - DSB-SC
  - SSB-SC or SSB)
  - VSB



DSBC-AM

- For DSBC – AM ,
  - k = 1, $C = V_c$ and $\phi_c = 0$

$$V_{am}(t) = [v_m(t) + V_c]\, cos(\omega_c t)$$

$$v_m(t) = V_m\, cos(\omega_m t)$$

- So

$$V_{am}(t) = V_c[1 + mcos\omega_m t]cos\omega_c t$$

$$m = \frac{V_m}{V_c}$$

DSBC-AM ../2

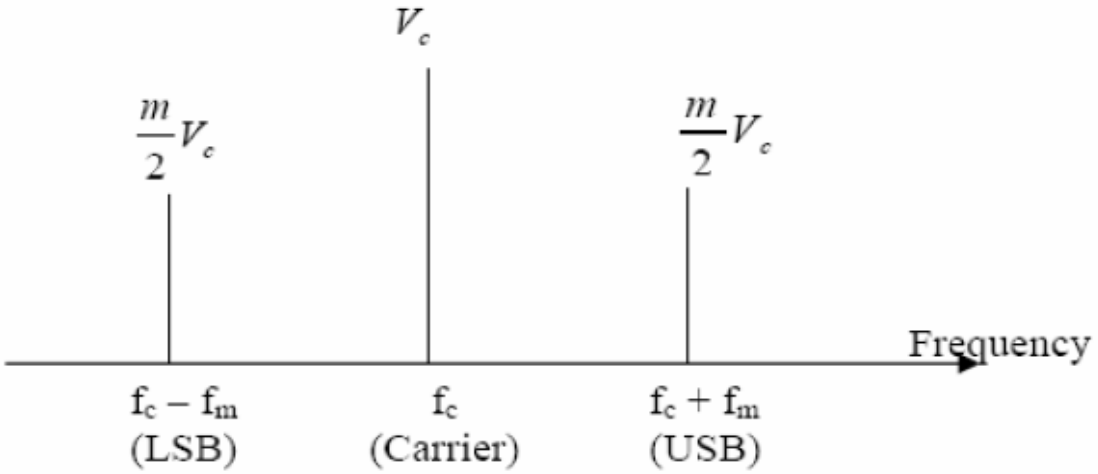
- So

$$v_{am}(t) = V_c \cos \omega_c t + \frac{mV_c}{2} \cos(\omega_c + \omega_m) t + \frac{mV_c}{2} \cos(\omega_c - \omega_m) t$$

- AM = sum of 3 sinusoids

DSBC-AM ../3

- DSBC spectrum



- Bandwidth of $V_{am}(t)$
  - $B = 2f_m$

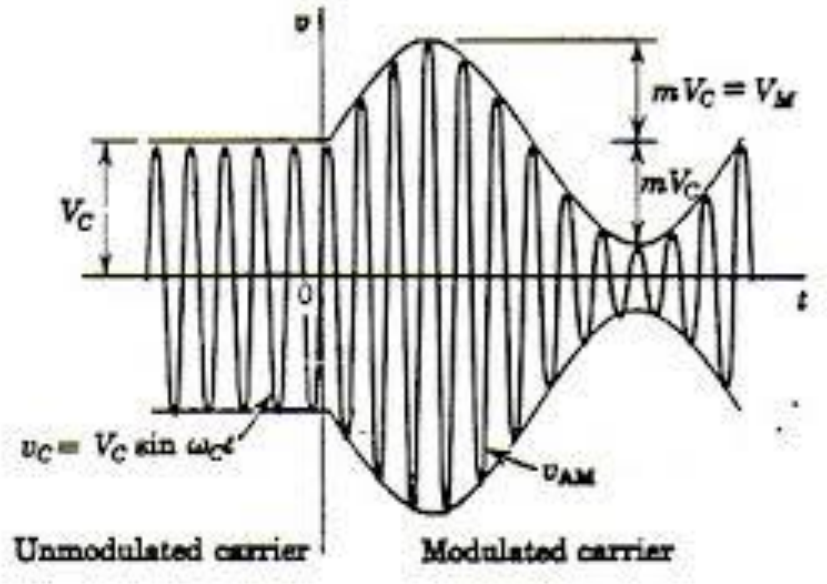
DSBC-AM ../4

- M*odulation index (*Percent modulation)

$m = V_m/V_c$ 0 < m < 1

$V_{max} = V_c + V_m$

$V_{min} = V_c - V_m$



- For sinusoidal modulation

$$m = \frac{V_{\max} - V_{\min}}{V_{\max} + V_{\min}}$$

Trigonometric Identies

- $\sin(A + B) = \sin A\cos B + \cos A\sin B$
- $\sin(A - B) = \sin A\cos B - \cos A\sin B$
- $\cos(A + B) = \cos A\cos B - \sin A\sin B$
- $\cos(A - B) = \cos A\cos B + \sin A\sin B$
- $\sin A\sin B = \frac{1}{2}[cos(A - B) - cos(A + B)]$
- $\cos A\cos B = \frac{1}{2}[cos(A - B) + cos(A + B)]$
- $\sin A\cos B = \frac{1}{2}[\sin(A + B) + \sin(A - B)]$
- $\cos A\sin B = \frac{1}{2}[\sin(A + B) - \sin(A - B)]$

- $\cos^2 A = \frac{1}{2}[1 + \cos 2A]$
- $\sin^2 A = \frac{1}{2}[1 - \cos 2A]$
- $\sin 2A = 2\sin A\cos A$
- $\cos 2A = \cos^2 A - \sin^2 A$
- $\sin^2 A + \cos^2 A = 1$

DSBC-AM ../5

- For various values of m

$$m = 0, \quad V_m = 0;$$

$$m = 1 = 100\%$$

$$m > 1$$

